Dilated Singer sets and the excess in Erdős's Sidon problem: a curve-counting theorem, forbidden-structure laws, a resonance theorem, exact Galois autocorrelations, and a reduction

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Abstract

Let $h(N)$ be the maximal size of a Sidon set in $\{1,\dots,N\}$. Erdős asked whether $h(N) = N^{1/2} + O(1)$ (Guy C9); the weaker $h(N) = N^{1/2} + O_\varepsilon(N^\varepsilon)$ is Erdős problem #30. We study random dilates of Singer perfect difference sets S ($|S| = q+1$, modulus $m = q^2+q+1$) and the number Z of points of a dilate in a window of length Cq. *Proved:* (i) if the factorial moments of Z converge to the Poisson ones for every fixed order and every window constant C, then $h(N) - N^{1/2}$ is unbounded (Proposition 2); (ii) for a ratio $\rho = u/v$, the number $N(\rho)$ of triples of S with $c-a = \rho(b-a)$ is the number of $\mathbb{F}_q$-points of a twisted form of a curve $C_{u,v}$ that does not depend on q, so $N(\rho)/E - 1 = O_\rho(q^{-1/2})$ whenever $C_{u,v}$ minus its diagonal is absolutely irreducible, which we certify by exact computation for 17 ratio orbits, including all non-forbidden ratios of height ≤ 4 (Theorem 1); (iii) the identity $\sum_\rho N(\rho)^2 = \sum_{\lambda,\mu}(|S \cap (\lambda S + \mu)|)_3$. *Observed, with pre-registered blind tests:* the deviation of the k-th factorial moment from the random value is, to first order, $-[\binom{k}{3}P_3 + \binom{k}{4}P_4]$, where P_3, P_4 are the window probabilities of 3-term progressions and parallelograms (asymptotically $-[\frac{3}{2}\binom{k}{3} + 2\binom{k}{4}]/(Cq)$), for $k \le 8$ (up to $k=7$ at $C=2$, $k=8$ at $C=4$) on 129 primes not previously used for $k \ge 4$; and $\sum_\rho (N(\rho)-E)^2$ equals the order of the multiplier group times $\sum_\rho E$, to within 0.8% for each of 37 primes $293 \le q \le 2969$ with m prime (mean 2.998), and within 10% for 21 of 23 prime powers $121 \le q \le 2197$. *New in Version 3:* $3k \mid N(\rho)$ (a free multiplier action, partly machine-checked in Lean), and a resonance theorem for cube $q = r^3$. For the six ratios of the S_3-orbit of $r + r^{-1}$ (roots modulo m of $x^3 - 3x + 1$ and $x^3 - 3x^2 + 1$), every point of the Singer line not fixed by the q-Frobenius has exactly r partners, so $N(\rho) = r(q+1-e)$ with $e \in \{0,1\}$ (m prime). This explains most of the excess of the variance ratio for cube q. There is one pre-registered blind test of the predicted excess ($q = 12167$), and pre-registered tests of the resonant orbit and of $M/\mu \approx r$ ($q = 4913, 6859, 15625$). The remaining step towards Erdős's question is a uniform estimate (an average over dilates), which we isolate but do not prove; the Erdős question remains open. *New in Version 4:* exact values of the Galois autocorrelation $C(l)$ of the underlying Jacobi sums at $l = 2, -1$ (minimal) and $l = 3, -2$: $C(l)$ is an explicit function of an elementary count $P(l)$ (Proposition 5), $P(2) = P(-1) = (q-2)^2$, $P(3) = (q-2)(2q-1)$ for $q \equiv 2 \pmod 3$, and $P(-2) = (q-2)(2q-1) + b_q$ for every prime $q \ge 5$, where $b_q = 0$ for $q \equiv 2 \pmod 3$ and $b_q = 4x^2 - 2q$ for $q = x^2 + 3y^2$ (numerically the q-th coefficient of the CM form $\eta(2\tau)^3\eta(6\tau)^3$) (Theorem 3; one step uses a known evaluation of a finite-field hypergeometric value, due to Ono and in the general form of Barman and Kalita).

1 Background

A set $A \subset \mathbb{Z}$ is Sidon if all sums $a + b$ ($a \le b$) are distinct. Erdős and Turán [4] showed $h(N) \le N^{1/2} + O(N^{1/4})$; Singer's construction [25] with the Baker–Harman–Pintz prime-gap theorem [1] gives $h(N) \ge N^{1/2} - O(N^{0.2625})$; the constant in the $N^{1/4}$ term has since been improved several times (Balogh–Füredi–Roy [2] and later work listed at [5]). Erdős offered USD 500 for deciding whether $h(N) = N^{1/2} + O(1)$ [7, 8]. The site [5] lists the weaker question $h(N) = N^{1/2} + O_\varepsilon(N^\varepsilon)$ as problem #30 (USD 1000), with the $O(1)$ form as a remark; the evidence below concerns the $O(1)$ question only. A Sidon set A modulo m with $|A| = k$, cut at its largest cyclic gap G, gives a Sidon set of integers in an interval of length $m - G$; for Bose sets [3] ($m = q^2 - 1$, $k = q$) this gives $k - \sqrt{N} \approx G/(2q)$. Martin and O'Bryant [8] note that the question would be answered (excess unbounded) if Bose sets had gaps that are not $O(q)$. Dilating A by a unit t preserves the Sidon property and moves the gaps; the use of dilates was suggested in B. Green's comment on [6] (see also [14]; earlier computer searches of dilates and cuts: [26, 10]), and Shakan [9] proved that every $A \subset \mathbb{F}_p$ with $|A| > 1$ has a dilate with a gap of at least $2p/|A| - 2$.

2 Dilate gaps of Bose sets

For every prime q in the ranges below we built Bose(q) $\subset \mathbb{Z}/(q^2 - 1)$, checked the Sidon property, and computed $G^*(q) = \max_t G(tA)$ over one multiplier per class $\{\pm t, \pm tq\}$. Positive control: $q = 829$ gives excess 10.2797, matching Zhang's bound $K(p_{145}) > 10.279$ ($p_{145} = 829$) [26].

- $101 \le q \le 2003$ (279 primes): a $\log q$ law fitted on $q \le 1000$, $g^* \approx -6.92 + 3.63 \ln q$, gives held-out ($q > 1000$) RMSE 1.22, against 4.25 for a constant fitted on the same data. (Saturating bounded models were not tested.)

- Blind test (law registered before computing): 48 primes $2011 \le q \le 5881$ (every 10th prime in $(2003, 6000]$): RMSE 1.52 (log) against 7.34 (constant), mean signed error -0.57. The largest q (5881) gives a Sidon set of size 5881 in $[1, 34{,}453{,}078]$ with excess 11.33; the largest excess in this set is 13.18 ($q = 5039$).

- Null model: random q-subsets of $\mathbb{Z}/(q^2 - 1)$ show the same growth (Bose slightly lower at 15 of 20 primes), so the $\log q$ growth is random-set behaviour; counts of 3-term relations $e_1 L_1 + e_2 L_2 + e_3 L_3 \equiv 0$ among the discrete logarithms are at the random level (mean per-q ratio 0.983 over 11 primes $101 \le q \le 383$; percentile bootstrap over the primes, 95% interval $[0.96, 1.00]$).

3 The moment route for Singer sets

Let S be a Singer set [25] ($|S| = n = q + 1$, $m = q^2 + q + 1$), $\ell = Cq$, and $Z_{t,s} = \#\{a \in S : (ta + s) \bmod m \in [0, \ell)\}$ for units t and shifts s. Let F_k be the average of the falling factorial $(Z)_k$ and $H_k = (n)_k(\ell)_k/(m)_k$ its value for a uniformly random n-subset; $R_k = F_k/H_k - 1$.

Proposition 1. *For every unit t, $\mathrm{Var}_s\, Z_{t,s} = q\ell(m - \ell)/m^2$ and $R_2 = 0$. If m is prime, averaging over all units,*

$$R_3 = \sum_{\rho \ne 0,1} \Big(\frac{N(\rho)}{E} - 1\Big) \frac{A(\rho)}{\sum_\sigma A(\sigma)}, \qquad N(\rho) = \#\{(a,b,c) \in S^3 \text{ distinct} : c - a \equiv \rho(b - a)\},\ E = \frac{(n)_3}{m - 2},$$

where $A(\rho)$ is the average over $u \in (\mathbb{Z}/m)^\times$ of the number of windows $[s, s+\ell)$ containing $\{0, u, u\rho\}$, and $\sum_\sigma A(\sigma) = \ell(\ell-1)(\ell-2)/(m-1)$.

(The first part holds because every nonzero residue is exactly one difference of tS and $(n)_2 = m-1$; for the second, m prime lets every triple be written as (a, b, c) with $c - a = \rho(b-a)$, and averaging over all units u sends it to every pair $(u, u\rho)$.) Since a perfect difference set has $N(-1) = N(2) = N(1/2) = 0$ and $A(\rho) \approx \ell^2/(2m)$ for these ρ, the three forbidden ratios contribute $-3/(2Cq)(1 + O(1/q))$ (within 0.2% for $q \geq 293$, 1.2% at $q = 41$).

- $R_k \sim c_k(C)/q$ (moments estimated from 4000 sampled multipliers): registered blind test for $k = 4$ on 11 new primes $409 \leq q \leq 601$ ($C = 2$: RMSE 0.0028 for c/q against 0.0238 for a constant; $C = 4$: 0.0013 against 0.0112); the same is seen descriptively for $k = 3, 5, 6$, with $|c_k|$ growing quickly in k.

- Exact over all units: registered tests on 18 primes $293 \leq q \leq 1373$ (m prime) and, from a sealed data set opened under a frozen plan, 17 primes $1559 \leq q \leq 2969$: the mean of $C\,qR_3$ is between -1.45 and -1.41, resp. between -1.47 and -1.46, inside the registered bands around $-3/2$ for all $C \in \{2, 3, 4, 5, 8\}$. The first test could not separate $-3/2$ from an earlier empirical value -1.17; the second band ($[-1.59, -1.31]$) excludes it.

- The RMS deviation D_{mid} of $N(\rho)/E$ over ratios of height 3–6: the registered decay test on $293 \leq q \leq 1373$ alone *failed* (slope -0.13 ± 0.17, outside the band $[-0.65, -0.35]$; few independent values per q). A second test, frozen before opening the sealed data but pooling them with the already seen first set, gives slope -0.504 ± 0.098 over $293 \leq q \leq 2969$ (the sealed set alone: -1.28 ± 0.39, i.e. noisy); $\sqrt{q}\, D_{\mathrm{mid}}$ lies between 0.49 and 1.84. Over all ratios the exponent is -0.50 in both sets, with $\sqrt{q}\, D_{\mathrm{all}} \approx 1.73$ for Singer sets against about 1.0 for random sets, consistent with counting noise inflated by the size-3 orbits of the multiplier q. These data are consistent with, but do not establish, a $q^{-1/2}$ law.

Sections 4–8 and 13 (new in Version 2; Sections 9–11 are new in Version 3) turn the heuristic of Version 1 into a theorem for each fixed ratio, give exact identities, explain the constants c_k, and state the reduction precisely.

4 The reduction

Proposition 2. *Fix $C > 0$ and let $\ell = \lfloor Cq \rfloor$. Suppose that for every fixed $k \geq 1$, $R_k = R_k(q, C) \to 0$ as $q \to \infty$ through primes. Then for all large q some dilate of S has an empty arc of length ℓ, and $\limsup_{N\to\infty}(h(N) - N^{1/2}) \geq (C+1)/2$. In particular, if this holds for every C, then $h(N) - N^{1/2}$ is unbounded.*

Proof. Choose (t, s) uniformly. Since $Z \leq n$, inclusion–exclusion gives $\Pr(Z = 0) = \sum_{k\geq 0}(-1)^k F_k/k!$, and the Bonferroni inequalities give $\Pr(Z = 0) \geq \sum_{k=0}^{2K+1}(-1)^k F_k/k!$ for every K. Since $H_k = (n)_k(\ell)_k/(m)_k \to C^k$, the hypothesis gives $F_k \to C^k$ for each fixed k. Choose K with $\left|\sum_{k\leq 2K+1}(-1)^k C^k/k! - e^{-C}\right| < e^{-C}/2$; then $\Pr(Z = 0) > 0$ for all large q, so some dilate tS misses a window of length ℓ. Rotating, tS lies in an interval of length $m - \ell$; a Sidon set modulo m is a Sidon set of integers, so we get $q+1$ elements in $[1, N]$ with $N = m - \ell = q^2 + (1-C)q + O(1)$ and $q + 1 - N^{1/2} = (C+1)/2 + O(1/q)$. □

The cutting step is classical (Section 1); what we add is the reduction of unbounded excess to the convergence of fixed-order factorial moments of window counts. We did not find it in the literature (Section 14).

5 Universal curves: a theorem for each fixed ratio

Write $S = \{i : g^i \in L\}$ with g a generator of $\mathbb{F}_{q^3}^\times$ and L a 2-dimensional $\mathbb{F}_q$-subspace. For $\rho = u/v$ put $(\alpha, \beta, \gamma) = (u - v, v, -u)$, so $\alpha + \beta + \gamma = 0$, and let

$$C_{u,v}: \qquad s^\alpha r^\beta t^\gamma = 1, \qquad (s-1)^\alpha (r-1)^\beta (t-1)^\gamma = 1.$$

Theorem 1. *(a) For q prime, $N(\rho)$ equals the number of $\mathbb{F}_q$-points of a twisted form of $C_{u,v}$ off its diagonal $s = r = t$, up to $O_\rho(1)$; the twist is Frobenius composed with the order-3 map $s \mapsto 1 - 1/s$. (b) For $\rho \in \{-1, 2, 1/2\}$, $C_{u,v}$ is the diagonal, so $N(\rho) = 0$. (c) If $C_{u,v}$ minus the diagonal is absolutely irreducible over $\overline{\mathbb{Q}}$, then $N(\rho)/E - 1 = O_\rho(q^{-1/2})$ for all primes $q > q_0(\rho)$. (d) The hypothesis of (c) holds for every ratio of height ≤ 4 other than $-1, 2, 1/2$ (five orbits) and for the orbits (under $\rho \mapsto 1 - \rho, 1/\rho$) of $-5, -5/2, -5/4, -6, -6/5, -7, -7/2, -7/6, -8, -8/3, -8/5, -8/7$: 17 orbits in all. With (b), 18 of the 22 orbits of height ≤ 8 are settled.*

Proof sketch. (a) With $x = g^a$, $y = g^b$, $z = g^c$ the condition is $x^{u-v} z^v y^{-u} \in \mathbb{F}_q^\times$. Parametrise $P(L) \cong \mathbb{P}^1(\mathbb{F}_q)$ by $x = e + sf$; the three conjugate linear forms $e^{q^i} + sf^{q^i}$ have distinct roots, and a Möbius change of variable over $\mathbb{F}_{q^3}$ sending them to $\infty, 0, 1$ turns $w \in \mathbb{F}_q^\times$, i.e. $w = w^q = w^{q^2}$, into the two equations of $C_{u,v}$ (the constants cancel because $\alpha + \beta + \gamma = 0$); Frobenius becomes the stated twist. (b) For exponents $(-2, 1, 1)$ the equations give $r + t = 2s$ and $rt = s^2$. (c) Absolute irreducibility over $\overline{\mathbb{Q}}$ persists modulo all large primes; apply the Weil bound for singular curves [11] to a p-independent projective model; excluded points are $O_\rho(1)$ and $E = q + O(1)$. (d) Eliminating one variable by a resultant gives a plane curve G; we exhibit smooth points of G over number fields whose degrees have $\gcd 1$ (or one quadratic or cubic field over which G stays irreducible). A geometric component through a smooth K-point is defined over a subfield of K, so the number of components divides every such degree and equals 1. Birationality of the projection is checked by subresultants. Exact arithmetic in SymPy; certificates and code are released. An independent review checked (a)–(c) and the certificate logic, and confirmed (a) by exact counts for $q = 5, 7, 11, 13$. □

Numerically, the genus of $C_{u,v}$ appears to grow at least quadratically in the height (for example $(n-3)(n-4)/2$ for exponents $(-n, 1, n-1)$, $n \le 6$, by Riemann–Hurwitz with numerical monodromy), so Theorem 1 alone does not give $R_3 \to 0$; see Section 13. Four further orbits $(-5/3, -7/3, -7/4, -7/5)$ exceeded our 10-hour certification budget; none failed.

6 Exact identities

Let $I_\lambda(\mu) = |S \cap (\lambda S + \mu)|$ and let m be prime. Two ordered triples of S with the same ratio determine a unique affine map $x \mapsto \lambda x + \mu$ taking one to the other, hence

$$\sum_{\rho \ne 0,1} N(\rho)^2 = \sum_{\lambda \in (\mathbb{Z}/m)^\times} \sum_{\mu} \big(I_\lambda(\mu)\big)_3,$$

which we also checked exactly for $q = 5, 17, 41, 59$. Since every nonzero residue is a unique difference, $\sum_\mu (I_\lambda(\mu))_2 = m - 1$ for every λ: second moments are exactly Poisson. For the multipliers $\lambda \in \{1, q, q^2\}$, λS is a translate of S and one μ gives $I = n$, contributing exactly $3(n)_3$. Under $x \mapsto x^\lambda$, a dilate by λ whose base-q digits have sum d maps the line L into a curve of degree d, so $I_\lambda \le d$ by Bézout unless L lies on that curve (this does not happen for $d = 2$ and odd q); for $d = 2$ (for example $\lambda = -1$, a conic) $(I_\lambda)_3 = 0$. Numerically, the non-multiplier λ with $\sum_\mu (I_\lambda(\mu))_3 = 0$

are exactly the nine elements $\{-1, 2, 1/2\} \cdot \{1, q, q^2\}$ for $q = 41, 59, 167, 173, 293, 383$ (all λ computed); the arc and oval facts behind -1 and 2 are classical [12, 13].

7 Higher moments: the forbidden-structure law (blind tests)

A Sidon set contains no 3-term progression and no non-trivial parallelogram $x + y = z + w$. Let $P_3(\ell)$, resp. $P_4(\ell)$, be the probability that an ordered distinct triple of positions in $[0, \ell)$ contains a 3-term progression, resp. that an ordered distinct quadruple is a parallelogram; $P_3 \sim 3/(2\ell)$ and $P_4 \sim 2/\ell$. The first-order prediction is

$$R_k \approx -\Big[\tbinom{k}{3} P_3(\ell) + \tbinom{k}{4} P_4(\ell)\Big] \sim -\frac{\frac{3}{2}\binom{k}{3} + 2\binom{k}{4}}{Cq}.$$

Two registered blind tests (blind for $k \geq 4$ in the first, for all k in the second) used plans frozen with a recorded SHA-256 before computing, 3000 sampled unit classes per q, and the inverse-variance pooled ratio measured/predicted. On 65 primes $607 \leq q \leq 1499$ with $C = 2$, $k = 3, \ldots, 6$ gave $1.02, 1.06, 1.06, 1.05$ (all 95% intervals contain 1). On the other 64 primes of that range, $C = 2$ gave $0.97, 0.97, 0.97, 0.97, 1.01$ for $k = 3, \ldots, 7$ (the $k = 8$ cell is void because its null control failed), and $C = 4$ gave $0.95, 0.98, 1.00, 1.00, 1.00, 1.01$ for $k = 3, \ldots, 8$. Random sets were within 3 standard errors of 0 in every other cell. So no new forbidden structure appears up to 8 points, and $c_k \sim k^4$ explains the fast growth noted in Version 1.

8 Variance law and prime powers

Conjecture 1. *$\sum_\rho (N(\rho) - E)^2 = |G| \sum_\rho E\,(1 + o(1))$, where $|G|$ is the order of the multiplier group ($|G| = 3$ for prime q, $|G| = 3k$ for $q = p^k$), the sums run over ρ with ρ and $\rho - 1$ units, and E is the mean of $N(\rho)$ over these ρ (equal to $(n)_3/(m-2)$ when m is prime).*

By Proposition 4 (Section 9), $3k \mid N(\rho)$ for every prime power $q = p^k$. For prime q with m prime, the ratio has mean 2.998 (s.d. 0.008, range $[2.979, 3.019]$) over 37 primes $293 \leq q \leq 2969$. For prime powers (Singer sets built in $\mathbb{F}_{p^{3k}}$; registered test) the ratio divided by $3k$ is given in Table 1. The per-λ errors $\sum_\mu (I_\lambda(\mu))_3$ minus the random value have mean about 0; excluding the nine null λ they are close to Gaussian, with standard deviation about $4.2\,q$ ($q = 383$: $4.21q$, kurtosis 3.01).

Proposition 3 (conditional)**.** *If $\sum_\rho (N(\rho) - E)^2 \ll q^3$, then $R_3 = O\big(\sqrt{\log q/q}\,\big)$.*

Proof sketch. By the Proposition of Section 3, $R_3 = \sum_\rho (N/E - 1)(w(\rho) - \bar{w})$ with $w = A/\sum A$, because $\sum_\rho N(\rho) = (n)_3$ makes the uniform part vanish. Cauchy–Schwarz gives $|R_3| \leq E^{-1}\big(\sum_\rho (N - E)^2\big)^{1/2} \|w - \bar{w}\|_2$, and lattice-point counting for $\{(x, y) : y \equiv \rho x\}$ in $(-\ell, \ell)^2$ gives $\|w - \bar{w}\|_2 \ll \sqrt{\log q}/q$. □

9 Orbits of the multiplier group

Proposition 4. *Let $q = p^k$ and let $N(\rho)$ count triples $(a, b, c) \in S^3$ with $b - a$ a unit and $c - a = \rho(b - a)$ (for prime m this is the count of Section 3). Then $3k \mid N(\rho)$ for every ρ.*

q	p^k	$3k$	good ρ	ratio	ratio/$3k$
121	11^2	6	2975	6.13	1.022
125	5^3	9	14059	9.91	1.101
128	2^7	21	11725	20.30	0.967
169	13^2	6	9145	6.53	1.089
243	3^5	15	50149	14.62	0.974
256	2^8	24	13145	24.35	1.014
289	17^2	6	16775	6.32	1.053
343	7^3	9	37135	10.62	1.180
361	19^2	6	30625	6.12	1.020
512	2^9	27	262655	28.04	1.039
529	23^2	6	55055	5.75	0.959
625	5^4	12	86855	12.05	1.004
729	3^6	18	449225	18.76	1.042
841	29^2	6	192335	6.00	1.000
961	31^2	6	195755	6.00	0.999
1024	2^{10}	30	245105	29.96	0.999
1331	11^3	9	1772891	9.48	1.053
1369	37^2	6	386425	6.07	1.012
1681	41^2	6	937945	6.00	1.001
1849	43^2	6	947903	5.98	0.996
2048	2^{11}	33	2997385	33.22	1.007
2187	3^7	21	4048957	21.03	1.001
2197	13^3	9	1609667	9.79	1.088

Table 1: Variance ratio $\sum(N-E)^2/\sum E$ over ρ with $\rho, \rho-1$ units, for Singer sets with $q = p^k$, $k \geq 2$ (registered test: within 10% of $3k$ for at least 80% of $q \geq 121$; here 21 of 23; the two misses and most of the excess at cube q are accounted for by the resonance of Section 10 (Table 2); for $q < 121$, outside the registered range, the ratio/$3k$ is far from 1, e.g. 0.20 at $q = 9$, 0.69 at $q = 49$, 0.83 at $q = 81$; later runs: all 11 later non-cube q $(2209, 2401, 2809, 3125, 3481, 3721, 4489, 6561, 8192, 14641, 16384)$ give ratio/$3k \in [0.9980, 1.0026]$; the later cube $q = 4096, 4913, 6859$ give $1.022, 1.038, 1.064$, see Table 2).

Proof. By the multiplier theorem, some translate of S is fixed by multiplication by p [15, 16]. N is translation invariant, so assume $pS = S$. We have $p^{3k} - 1 = (q-1)m$. If $p^j \equiv 1 \pmod{m}$ with $0 < j < 3k$, then $m \mid p^d - 1$ for $d = \gcd(j, 3k) \leq 2k$, which is impossible because $0 < p^d - 1 < m$. So p has order $3k$ modulo m. The map $(a, b, c) \mapsto (pa, pb, pc)$ preserves the set of triples. A triple fixed by p^j, $0 < j < 3k$, would give $(p^j - 1)(b - a) = 0$ with $b - a$ a unit, which is impossible. So the action is free and every orbit has $3k$ elements. □

This is elementary and probably known. The order statement and the freeness step are machine-checked in Lean 4 with Mathlib (files `C9Order.lean`, `C9Orbit.lean`; standard axioms only). The multiplier theorem itself is not formalised. So Conjecture 1 says that $M = N/3k$ is Poisson-like. This holds for non-cube q (Table 1) but fails for cube q; the next two sections explain most of this failure.

10 A resonance theorem for cube q

Let $q = r^3$, where r is a prime power. Let $F = \mathbb{F}_q$ and $K = \mathbb{F}_{q^3} = \mathbb{F}_{r^9}$. Write $\tau(u) = u^r$ and $\varphi = \tau^3$ (the q-Frobenius). Let $L = \ker \mathrm{Tr}_{K/F}$, so that $S = \{i : g^i \in L\}$. Since τ commutes with φ, we have $\tau(L) = L$. Moreover φ is F-linear on L with $\varphi^3 = 1$. Since $r^9 = q^3 \equiv 1$ and $r^3 = q \not\equiv 1 \pmod m$, r has order 9 modulo m. Put $\rho_1 = r + r^{-1} \bmod m$.

Theorem 2. *Let $x \in P(L)$ be a point of the Singer line.*

- *If $\varphi x \neq x$ (projectively), then there are exactly r points $y \in P(L) \setminus \{x\}$ with $z/x = (y/x)^{\rho_1}$ for some $z \in P(L)$.*
- *If $\varphi x = x$, there are none.*

The number e of φ-fixed points of $P(L)$ is 0, 1 or 2 according as $r \equiv 2$, 0 or 1 (mod 3). Consequently, if m is prime, $N(\rho) = r\,(q + 1 - e)$ for each of the six ratios ρ in the S_3-orbit of ρ_1; this orbit contains $\rho_0 = r^4 + r^{-4} = (r + 1)r^4$. If $r \equiv 1 \pmod 3$ then $3 \mid m$, so for prime m we have $e \in \{0, 1\}$, and $e = 1$ exactly when $p = 3$.

Proof. (i) *Reduction.* Put $u = y/x$. Then $y \in P(L)$ iff $\mathrm{Tr}(xu) = 0$. If x is not φ-fixed, then $\psi = \varphi(x)/x \notin F$ and $\mathrm{Tr}(x\psi) = \mathrm{Tr}(\varphi x) = 0$, so $\{u : \mathrm{Tr}(xu) = 0\} = F + F\psi$. In $K^\times/F^\times$ the exponent ρ_1 acts as $u \mapsto \tau(u)\tau^{-1}(u)$, so we need $z = x\,\tau(u)\,\tau^{-1}(u) \in F^\times L$. That is, $\mathrm{Tr}(x\,\tau(u)\,\tau^{-1}(u)) = 0$. Take $u = 1 + s\psi$ with $s \in F$; on F we have $\tau s = s^r$ and $\tau^{-1}s = s^{r^2}$. Using $\mathrm{Tr}(x) = 0$, the condition reads $As^r + Bs^{r^2} + Cs^{r+r^2} = 0$, where $A = \mathrm{Tr}(x\tau\psi)$, $B = \mathrm{Tr}(x\tau^{-1}\psi)$ and $C = \mathrm{Tr}(x\,\tau\psi\,\tau^{-1}\psi)$ lie in F. Remove $s = 0$ (that is, $y = x$) and put $Z = s^{-r}$, with $s = \infty$ corresponding to $Z = 0$. This gives the affine $\mathbb{F}_r$-linear equation

$$AZ^r + BZ + C = 0, \qquad Z \in F.$$

(ii) *Two identities.* Write $\xi_j = \tau^j(x)$ for j mod 9, with $\xi_j + \xi_{j+3} + \xi_{j+6} = 0$. Then A, B, C are rational functions of $\xi_0, \ldots, \xi_5$ with integer coefficients and monomial denominators, and τ acts as the index shift. Let D be the Dickson matrix of $f(Z) = AZ^r + BZ$. Exact computation in Sage gives

$$N_{F/\mathbb{F}_r}(A) + N_{F/\mathbb{F}_r}(B) = 0 \quad \text{and} \quad \mathrm{adj}(D)\,(C, C^r, C^{r^2})^T = 0.$$

After clearing the (nonzero) denominators, both are identities between polynomials with integer coefficients that hold over $\mathbb{Q}$. They therefore hold in every characteristic. They were also rechecked directly over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$. (iii) *Non-degeneracy.* The numerator of A factors as $(\xi_1\xi_3 - \xi_0\xi_4)(\xi_1^2 + \xi_1\xi_4 + \xi_4^2)$, and the denominator is $-\xi_1\xi_4\xi_7 \neq 0$. The second factor vanishes iff ψ is a root of $t^2 + t + 1$. The first vanishes iff $\tau x = cx$ with $c \in F$, which gives $\varphi x = N_{F/\mathbb{F}_r}(c)\,x$. In both cases x is φ-fixed. So $A \neq 0$, hence $N(B) = -N(A) \neq 0$ and $\mathrm{rank}\, D = 2$. (iv) *Counting.* $N(-B/A) = N(-1)\,N(B)/N(A) = (-1)^{1+r+r^2} \cdot (-1) = 1$, so the kernel of f has exactly r elements and $|\mathrm{Im}\, f| = r^2$. Since $\mathrm{adj}(D)\,D = \det(D)\,I = 0$, each row (w_0, w_1, w_2) of $\mathrm{adj}(D)$ annihilates (b, b^r, b^{r^2}) for every $b \in \mathrm{Im}\, f$. A nonzero row gives a nonzero linearised polynomial of degree at most r^2. Its zero set therefore has at most r^2 elements and contains $\mathrm{Im}\, f$, so it equals $\mathrm{Im}\, f$. By (ii), $-C \in \mathrm{Im}\, f$, so there are exactly r solutions Z, that is, r partners y. (v) *Fixed points.* The φ-fixed points of $P(L)$ are the eigenlines of the F-linear map $\varphi|_L$, whose eigenvalues are cube roots of unity in F. The eigenvalue 1 needs $x \in F$, and then $\mathrm{Tr}(x) = 3x$ vanishes only if $p = 3$.

- If $r \equiv 2 \pmod 3$, then F has no primitive cube root of unity, so $e = 0$.
- If $p = 3$, then $\varphi|_L$ is unipotent with fixed line $F \cdot 1$, so $e = 1$.

- If $r \equiv 1 \pmod 3$, then $\varphi|_L$ has the two eigenvalues ω, ω^2 ($\omega \in F$ a primitive cube root of unity), so $e = 2$.

For a fixed point:

- If $\varphi x = \omega x$, then K splits into φ-eigenlines, which τ preserves, so $\tau x = cx$ and $\tau^{-1}x = c'x$ with $c, c' \in F$, and $\{u : \mathrm{Tr}(xu) = 0\} = F + Fx$. For $u = \beta_0 + \beta x$ the condition becomes $3cc'\beta^{r+r^2}x^3 = 0$, i.e. $\beta = 0$, i.e. $y = x$.

- If $p = 3$, then $L = F + Fv$ with $\varphi v = v + \alpha$, $\alpha \in F^\times$. For $y = \beta_0 + \beta v$, the condition reduces to $\mathrm{Tr}(v^2)\,\beta^{r+r^2} = 0$ with $\mathrm{Tr}(v^2) = -\alpha^2 \neq 0$, so again $y = x$.

Summing over x gives $N(\rho_1)$ when m is prime, because then every nonzero difference is a unit. The maps $(a, b, c) \mapsto (b, a, c)$ and $(a, b, c) \mapsto (a, c, b)$ realise $\rho \mapsto 1 - \rho$ and $\rho \mapsto 1/\rho$ and preserve the unit conditions, so N is constant on the S_3-orbit. □

The theorem was also checked by independent brute force for $r \leq 13$; for $q = 8, 27, 512, 1331$ it was checked on all six ratios. An independent AI review accepted it after the fixes included here.

Algebraic form. The six ratios of the orbit are roots modulo m of $x^3 - 3x + 1$ and $x^3 - 3x^2 + 1$. For prime m they are all of the roots; for composite m these cubics can have more roots, by the Chinese remainder theorem. The first is the minimal polynomial of $2\cos(2\pi/9) = \zeta_9 + \zeta_9^{-1}$, with $\zeta_9 \mapsto r$. Its root set is invariant under $x \mapsto 1/(1 - x)$, the inverse of the order-3 map $s \mapsto 1 - 1/s$ of Theorem 1(a).

11 The resonance and the cube excess

For the six resonant ratios $M(\rho) \approx r\mu$. If all other ratios behaved as in Conjecture 1, then $\mathrm{Var}\, M/\mathbb{E}\, M$ would exceed 1 by about

$$\Delta = 6(r-1)^2\mu/n_{\mathrm{good}}, \qquad n_{\mathrm{good}} = \prod_{\ell^e \| m} \ell^{e-1}(\ell - 2), \quad \mu = \frac{(q+1)q(q-1)}{3k\, n_{\mathrm{good}}} \prod_{\ell \mid m} \frac{(\ell-1)(\ell-2)}{\ell^2}.$$

This is a heuristic, not a theorem; it decays like $q^{-1/3}$.

Registration.

- The value of Δ was registered before computation only for $q = 12167$.

- The statements "the six largest $M(\rho)$ are roots of the cubics" and "$M/\mu \in [0.8r, 1.2r]$" were registered before computation for $q = 4913, 6859$. For $q = 15625$ the registration was "the cubic orbit is among the twelve largest, with $M/\mu \in [20, 30]$".

- All other entries of Table 2 are retrospective.

The plan file and its SHA-256 ledger are released. Registration times are self-attested.

For $k = 3$ and $q \geq 1331$, the formula gives 94–100% of the observed excess. In the other rows of Table 2 it gives 55–100% (lowest at $q = 512$). Two earlier registered readings failed at $q = 6859$: a $q^{-1/2}$ decay band, and a $p \equiv 1 \pmod 3$ effect. Registered tests at $q = 24389$ and $q = 32768$ are pending. For $k = 6, 9, 12$ a second, weaker orbit appears, with $M/\mu = 2.77$, 3.91, 4.00, 2.98 at $q = 729, 4096, 15625, 19683$. Its ratios are not roots of integer polynomials of degree ≤ 4 with coefficients bounded by 4, and they are not of the form $p^a \pm p^b \pm p^c$. We have not identified this orbit. At $q = 19683$ it contributes only about 10% of the residual 0.0015; most of the residual comes from several hundred ratios with $M/\mu \approx 2$. Weaker levels with $M/\mu \approx 2$ also occur for $k = 3$ ($q = 4913, 12167$) and for the non-cube control $q = 16384$. They are also unexplained.

q	k	Δ formula	observed	top-6 M/μ (r)	status
125	3	0.089	0.101	4.91 (5)	retrospective
343	3	0.143	0.180	7.19 (7)	retrospective
512	9	0.021	0.039	8.03 (8)	retrospective
729	6	0.032	0.042	8.86 (9)	retrospective
1331	3	0.0500	0.0534	11.0 (11)	retrospective (local computation)
2197	3	0.0873	0.0877	– (13)	retrospective
4096	12	0.020	0.022	15.93 (16)	retrospective
4913	3	0.0368	0.0377	17.01 (17)	top-6, M/μ registered
6859	3	0.0631	0.0635	18.94 (19)	top-6, M/μ registered; Δ retrospective
12167	3	0.0281	0.0281	22.99 (23)	**Δ registered (blind)**; top-6, M/μ retrospective
15625	6	0.0261	0.0261	24.95 (25)	top-6, M/μ registered; Δ not registered
19683	9	0.0077	0.0092	27.01 (27)	exploratory ($p = 3$)

Table 2: The resonance and the cube excess. Wherever the six largest $M(\rho)$ were recorded, they are roots of the two cubics and $M/\mu \approx r$. Non-cube controls ($q = 8192, 14641, 16384$) give ratio$/3k \in [0.9997, 1.0024]$.

12 Galois autocorrelation of Jacobi sums: exact values

Let $K = \mathbb{F}_{q^3}$ (q an odd prime), $m = q^2 + q + 1$, and let $L : K^\times \to \mathbb{Z}/m$ be the discrete logarithm modulo m; its kernel is $\mathbb{F}_q^\times$. Let ψ be a character of $K^\times$ of order m (trivial on $\mathbb{F}_q^\times$) and put $J[a,b] = \sum_{y \in K \setminus \{0,1\}} \psi^a(1-y)\psi^b(y)$, a Jacobi sum. For a unit l modulo m define the *Galois autocorrelation*

$$C(l) = \sum_{a \neq 0} \sum_{b} J[a,b]\,\overline{J[la, lb]},$$

i.e. the correlation of the Jacobi sums with their conjugates under $\zeta_m \mapsto \zeta_m^l$. Correlations of this kind are the missing input of Section 13. We also use the elementary count

$$P(l) = \#\{(y,c,d) :\ y \in K \setminus \{0,1\},\ c,d \in \mathbb{F}_q^\times,\ c\,y^l + d\,(1-y)^l = 1\}$$

(equivalently, pairs y, y' with $L(1-y') = l\,L(1-y)$ and $L(y') = l\,L(y)$).

Proposition 5. *$C(l) = m^2 P(l) - ((q-1)m-1)^2 - (m-1)$ for every unit l, and $P(l) = P(l^{-1})$.*

Proof. By orthogonality, $\sum_{a,b} J[a,b]\overline{J[la,lb]} = m^2 \#\{(y,y') : L(1-y) = lL(1-y'),\ L(y) = lL(y')\} = m^2 P(l^{-1})$, and swapping y, y' gives $P(l^{-1}) = P(l)$. The row $a = 0$ equals $\sum_b J[0,b]\overline{J[0,lb]}$ with $J[0,0] = q^3 - 2 = (q-1)m - 1$ and $J[0,b] = -1$ for $b \neq 0$. □

For $y \in \mathbb{F}_q \setminus \{0,1\}$ there are exactly $q-2$ admissible pairs (c,d), so $P(l) \geq (q-2)^2$ always. For $l = 3$ and $l = -2$ the equation is a polynomial over $\mathbb{F}_q$ of degree ≤ 4 in y, and its roots in $K \setminus \mathbb{F}_q$ come from irreducible cubic factors; hence $P(l) = (q-2)^2 + 3I(l)$, where $I(l)$ is the number of $(c,d) \in (\mathbb{F}_q^\times)^2$ for which the polynomial has an irreducible cubic factor (I is not the triple count $N(\rho)$ of the earlier sections).

Theorem 3. *Let $q \geq 5$ be prime. Define $b_q = 0$ if $q \equiv 2 \pmod 3$ and $b_q = 4x^2 - 2q$ if $q = x^2 + 3y^2$ ($q \equiv 1 \pmod 3$). (Numerically, b_q is the q-th coefficient of $\eta(2\tau)^3\eta(6\tau)^3$; this is not used below.)*

(a) $P(2) = P(-1) = (q-2)^2$, the least possible value.

(b) If $q \equiv 2 \pmod 3$, then $P(3) = (q-2)(2q-1)$.

(c) $I(-2) = \big((q-2)(q+1)+b_q\big)/3$, *hence* $P(-2) = (q-2)(2q-1)+b_q$. *In particular* $P(3) = P(-2)$ *when* $q \equiv 2 \pmod 3$.

Proof of (a). For $l = 2$ and $l = -1$ (after clearing denominators) the equation is a nonzero polynomial of degree ≤ 2 over $\mathbb{F}_q$. Its roots lie in $\mathbb{F}_{q^2} \cap K = \mathbb{F}_q$, so only the $(q-2)^2$ solutions with $y \in \mathbb{F}_q$ remain. □

Proof of (b), sketch. Write $c\,y^3 + d(1-y)^3 - 1$ as $(c-d)\,g(y)$ with $g(y) = y^3 + b(y^2-y) + e$. The map $(c,d) \mapsto (b,e)$ is a bijection onto the pairs with $b \notin \{0,-3\}$ and $b \ne 3e$. On the excluded lines, g has a root in $\mathbb{F}_q$, since cubing is bijective on $\mathbb{F}_q$. In the family $x^3+b(x^2-x)+e$, two distinct roots determine the third, except when the two roots sum to 1, which needs $a^2 - a + 1 = 0$; this has no solution because -3 is a non-square. Counting triple roots (2), double roots $(q-3)$, ordered pairs of roots and root incidences gives $(q^2-4q+7)/6$ split, $(q^2+1)/2$ one-root and $(q-2)(q+1)/3$ irreducible members. Hence $I(3) = (q-2)(q+1)/3$ and $P(3) = (q-2)^2 + 3I(3)$. □

Proof of (c), outline. The full proof is in the release (file `N2_PROOF.md`); we list the steps. Let ϕ be the quadratic character of $\mathbb{F}_q$. (1) For distinct roots $y_1, y_2 \in \mathbb{F}_q \setminus \{0,1\}$ the two root conditions are linear in (c,d), with determinant $(y_2-y_1)(y_1+y_2-2y_1y_2)$, $c = y_1^2y_2^2(y_2-y_1)(2-y_1-y_2)/\det$ and $d = (1-y_1)^2(1-y_2)^2(y_2^2-y_1^2)/\det$. Put $u = y_1+y_2$ and $v = y_1y_2$. (2) The number of admissible unordered pairs is $P_2 = (q-3)(q-5)/2$. (3) The residual quadratic of the quartic has discriminant $u(u-2)(2v-u+2)/(2v-u)$. Summing its quadratic character over admissible pairs gives $S_2 = (T-4q+14)/2$ with $T = \sum_{x,z} \phi\big((x^2-1)(z^2-1)(x^2+1-2z)\big)$. The excluded lines contribute $2q-4$, $2q-6$, -3 and 0. (4) A change of variables gives $T = 1 + \phi(-1)G(4)$, where $G(\lambda) = \sum_{x,y} \phi\big(xy(1-x)(1-y)(x-\lambda y)\big)$, and $q^2\,{}_3F_2(4) = \phi(-1)G(4)$ for Greene's ${}_3F_2$ with quadratic characters. (5) The repeated-root types have counts $E_1 = (q+1-z_0)/2$ (type $(1)^2(2)$), $E_2 = (q-5-z_0)/2$ (type $(1)(1)(1)^2$) and $E_3 = z_0/2$ (type $(1)^2(1)^2$), where $z_0 = 1+\phi(-3)$. A double root a forces $c = a^3$ and $d = -(a-1)^3$. (6) Counting roots, pairs and character-weighted pairs by factorisation type gives $I(-2) = (q-2)^2 - E_1 - \frac{4}{3}P_2 + \frac{2}{3}S_2 - \frac{1}{3}E_2 - \frac{4}{3}E_3$. (7) The value $q^2\,{}_3F_2(4) = q + b_q$ follows from [17, Thm. 1.7] (Greene's ${}_3F_2$ [20], character $S = \phi$, allowed there since ϕ has order 2); the authors of [17] state that $S = \phi$ gives a result of Ono [24]. For $q \equiv 2 \pmod 3$ the value is $1/q$. For $q \equiv 1 \pmod 3$ the theorem gives $[(\phi\chi_3 \text{ over } \chi_3) + (\phi\chi_3^2 \text{ over } \chi_3^2)]^2 - 1/q$; here the Hasse–Davenport relation [21] gives $J(\phi\chi_3, \bar\chi_3) = \chi_3(2)\overline{J(\chi_3,\chi_3)}$, and the cubic character of 2 [21] makes the real part $\pm x$, so $q^2\,{}_3F_2(4) = 4x^2 - q = q + b_q$. Substituting (2)–(7) into (6) gives (c). □

Verification. Each step was checked by exact computation:

- steps (1)–(6) on all primes $5 \le q \le 31$;
- step (7) on $7 \le q \le 97$;
- the final formula for $I(-2)$ on all primes $5 \le q \le 53$, by an independent brute-force count;
- the identity $b_q = a_q(\eta(2\tau)^3\eta(6\tau)^3)$ on all primes $5 \le q \le 199$ (numerical only).

The symbolic identities were checked with SymPy. An independent AI review (local, with its own code) found no error; it is not a human expert review.

Geometric explanation (not used in the proof). We identified the source of b_q by exact computation. In the twisted form, the level sets of $\mathrm{Tr}(\alpha)\mathrm{Tr}(\alpha^{-1})$ are the members of the pencil $(x+y)(y+z)(z+x) = k\,xyz$. This pencil is the extremal rational elliptic surface with singular fibres I_6, I_3, I_2, I_1, i.e. the universal curve for $\Gamma_1(6)$. The double cover branched at the I_3 and I_1

fibres is the singular K3 surface with fibres $3I_6 + 3I_2$ and transcendental lattice $\mathrm{diag}(2, 6)$ [19], which has complex multiplication by $\mathbb{Q}(\sqrt{-3})$; this is why the error term is a CM coefficient.

Observations.

- $P(4) = P(-2)$ for all 20 primes $5 \le q \le 79$ tested. Besides $P(ql) = P(l)$ (Frobenius) and $P(l^{-1}) = P(l)$, a search over Möbius maps with integer coefficients in $[-3, 3]$ found no further symmetry of P ($q = 11, \dots, 23$), so this equality is not explained by such a map.
- Among the rationals a/b with $|a|, |b| \le 12$ that are units modulo m for every tested q ($11 \le q \le 41$; this excludes all a/b involving 3 or 7), the only classes of equal P beyond $\{l, l^{-1}\}$ are $\{2, -1, \frac{1}{2}\}$ and $\{-2, -\frac{1}{2}, 4, \frac{1}{4}\}$. For $q \equiv 2 \pmod 3$ the second class also contains 3 and $\frac{1}{3}$, by Theorem 3(b),(c).
- For other small heights, $(P(l) - (q-2)^2 - q^2)/q$ fluctuates more as the height grows: over the primes $47 \le q \le 79$ its standard deviation grows from about 1 at $l = -2$ to about 4 at $l = 7$ (smaller q include multiplier coincidences such as $l = q$, which inflate it). This is consistent with cohomology that grows with the height, so these exact formulas do not by themselves give the averaged bound of Section 13.
- $N(\rho)$ is an average of root counts of the trinomials $c\,y^\rho + y - 1$. Kelley proved, under GRH, that the root counts of random-coefficient trinomials over prime fields are Poisson, and conjectured the same for a fixed prime [22, 23, 18]. The Poisson-type laws of Sections 4–9 are an extension-field analogue in a different regime.

13 What remains

By Proposition 2, a negative answer to the $O(1)$ question follows from $R_k \to 0$ for each fixed k and each C. For $k = 3$ this follows from Conjecture 1, or from any bound $o(q^4/\log q)$ on the variance. For $k \ge 4$ a different decomposition is needed. In Fourier terms the missing input is cancellation, on average over the dilation parameter, in correlations of Jacobi sums over $\mathbb{F}_{q^3}$, beyond the square-root (Weil–Deligne) bound for single sums. We do not know how to prove it. Section 12 evaluates these correlations exactly at a few small dilations. Three cheap routes to the averaged bound failed in our tests: the exact formulas grow in complexity with the height of the dilation; among small heights (numerators and denominators ≤ 12, avoiding 3 and 7) we found no further coincidences of P beyond those explained by Theorem 3 and one unexplained class; and maximum bounds for roots of trinomials [18] are far too weak for an L^2 statement.

14 Limitations, literature and disclosure

Theorem 1 covers fixed ratios only (prime q, or large characteristic). The higher-moment and variance laws are numerical. Theorem 2 is proved; two polynomial identities and one factorisation in its proof were verified by exact computer algebra (Sage 10.7; the scripts and their SHA-256 hashes are released) rather than by hand. The excess formula of Section 11 is a heuristic, not a theorem. For Version 3, a search (web, zbMATH Open, three full texts; 20 October 2026) found no prior statement of Theorem 2. Proposition 4 is elementary and probably known. Proposition 2 is conditional on moment convergence, which is not proved. A full-text literature review and a zbMATH Open search (19 October 2026) found the forbidden ratios, the arc and oval facts [12, 13] and the dilate strategy ([6], Shakan [9]). They did not find the curve count, the two laws,

the identity of Section 6 or the reduction of Section 4. Guy's C9 entry and MathSciNet were not checked. Computations and drafting were done with AI assistance (Anthropic Claude). The statements labelled as proved were checked by an independent AI review and by exact computation, but not yet by a human expert. For Version 4 (searches on 30 October 2026: zbMATH Open, web, GitHub), we found no prior statement of Theorem 3 or of Proposition 5. Also not checked for Version 4: the original paper of Jungnickel and Vedder, the full text of the paper of Ono, and a 2005 essay on Singer sets (A. Chmelnitzki), whose server was unreachable. Theorem 3(c) relies on [17, Thm. 1.7] (statement read; a preprint) and on standard facts from [21].

Data and code

Version 1 release: `https://github.com/vichienFF/ramanujan-explorer/tree/main/erdos_sidon`. Version 2 additions (certificates, blind-test plans with SHA-256 ledger, Kaggle outputs): `https://github.com/vichienFF/ramanujan-explorer/tree/main/erdos_sidon/v2`. Version 3 additions (Lean files, Sage scripts for Theorem 2, resonance blind tests): `https://github.com/vichienFF/ramanujan-explorer/tree/main/erdos_sidon/v3`. Version 4 additions (proof of Theorem 3, verification scripts, independent review): `https://github.com/vichienFF/ramanujan-explorer/tree/main/erdos_sidon/v4`.

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